



B-Tree Insertion & Splitting

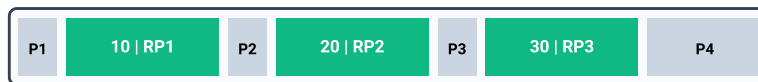
Gate Smashers Revision Notes & Step-by-Step Numerical Guide

1. Order-4 B-Tree Properties

For order $P = 4$, the maximum children capacity is 4. The table below outlines key upper and lower bounds essential for solving numerical problems:

Property	Formula (Order P)	Order P = 4 Value
Maximum Children	P	4
Maximum Keys	$P - 1$	3
Minimum Keys	$\lceil P / 2 \rceil - 1$	$\lceil 4/2 \rceil - 1 = 1$
Maximum Block Pointers	P	4
Maximum Record Pointers	$P - 1$	3

2. B-Tree Node Layout (Order 4)



3. What is Overflow & How to Resolve It?

For order 4, the maximum allowed keys per node is 3. Inserting a 4th key triggers an **overflow** condition, resolved via 3 steps:

Step 1: Find Left Median

For keys [1 | 2 | 3 | 4], candidate medians are 2 and 3. Standard convention chooses **Left Median = 2**.

Step 2 & 3: Promote & Split

Promote **2** to the parent. Split remaining keys into left child [1] and right child [3 | 4].

4. Step-by-Step Insertion Trace (Keys: 1 to 10)

Insert 1, 2, 3 (No Overflow)

Root becomes [1 | 2 | 3]. Valid because maximum capacity = 3 keys.

Insert 4 → Overflow & First Split

Temporary state: [1 | 2 | 3 | 4] (Overflow!). Left Median = 2 promoted upward.



5. Continuation: Tracing Inserts 5 through 10

Insert 5 & 6 (Right Child Split)

5 enters right leaf [3 | 4 | 5]. Inserting 6 yields [3 | 4 | 5 | 6] (Overflow!). Promote median **4** to root.



Insert 7 & 8 (Root Expansion)

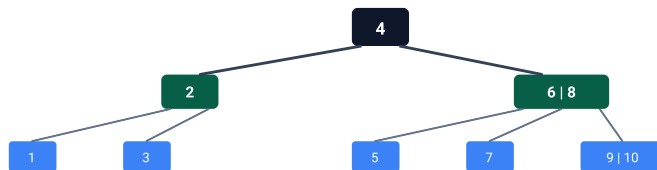
7 enters right leaf [5 | 6 | 7]. Inserting 8 causes overflow [5 | 6 | 7 | 8]. Promote median **6** to root.



Insert 9 & 10 → Cascading Root Split ★

9 enters rightmost leaf [7 | 8 | 9]. Inserting 10 creates [7 | 8 | 9 | 10]. Promote median **8** to root. Root temporary state becomes [2 | 4 | 6 | 8] (Root Overflow!). Left median of root = **4** is promoted to create a new top-level root.

6. Final B-Tree Structure



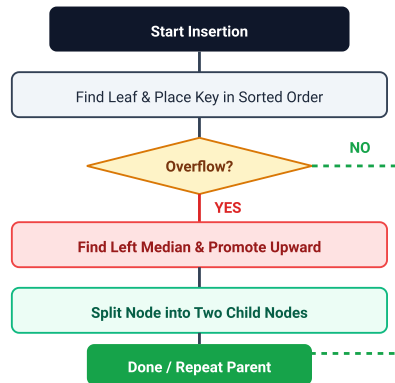
Total Levels
3 Levels

(Level 0, Level 1, Level 2)

Tree Height
Height = 2

(Last Level - Root Level)

7. B-Tree Insertion Algorithm Flowchart



! Exam Traps to Avoid

- **Trap 1:** Order $P = 4$ means max 4 *children*, NOT 4 keys. Max keys = $P - 1 = 3$.
- **Trap 2:** Never drop overflow keys. Always find median → promote → split.
- **Trap 3:** Promotion can cause cascading splits up to the root, increasing tree height.
- **Trap 4:** Level vs Height – Height = Total Levels – 1 (when root is Level 0).

🔑 10-Second Revision Pattern

Insert → Sort → Overflow Check → Left Median → Promote → Split → Repeat Upward